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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ACTIVE LEARNING METHODOLOGY REPORT

ADAPTIVE AR WORK INSTRUCTIONS FOR MACHINE ASSEMBLY LINES

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Assessment

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ADAPTIVE AR WORK INSTRUCTIONS FOR MACHINE ASSEMBLY LINES

Abstract

Assembly workers on a machine assembly line follow paper manuals or fixed screens, so they must look away from the part, search for the right page, and cope with the same level of detail whether they are new or highly experienced. This report describes an Adaptive Augmented Reality (AR) work instruction system for a gearbox assembly line. An AR device (tablet or head-mounted display) recognises the machine part with the camera and overlays the next step directly on it: arrows, highlighted bolts, a ghost image of the part to be fitted, the torque value and a short text. The system adapts to the operator. A skill score calculated from step time and errors decides whether the instruction is shown in full detail with animation (novice), with key values only (expert), or in between. Sensors and a vision checker confirm every step, lock the tool when a wrong part is used, and send the data to a supervisor dashboard.

Objectives

- To show assembly steps in the worker's own view so that the eyes and hands stay on the product.
- To adapt the amount of guidance automatically to the skill and the performance of each operator.
- To detect wrong parts, wrong torque and skipped steps while they happen and prevent defective units.
- To reduce training time, cycle time and assembly errors, and to give the supervisor live data.

Tools and technologies

Component / Tool	Specification	Purpose
AR device	Android tablet / HoloLens 2 class headset	Camera, display, voice commands
Unity + AR Foundation / Vuforia	Unity 2022 LTS, Model Targets from CAD	Tracking and 3D overlay
Vision checker	Lightweight CNN (part, bolt, orientation)	Detect wrong or missing parts
IoT torque tool	MQTT messages over the plant Wi-Fi	Confirm tightening torque
Web studio (HTML5 / JavaScript)	Browser based authoring and dashboard	Create instructions and view data
MES / PLC	Work order and station signals	Link the system with the line

Key idea in one line

Instruction detail = f (operator skill score, step time, error count, step risk). The more help a worker needs, the more the AR overlay shows; the more skilled the worker, the more the overlay stays out of the way.

1. INTRODUCTION

1.1 Background

Machine assembly lines build products such as gearboxes, engines and electrical panels from many parts. Operators follow work instructions that tell them which part to pick, where to fit it and how tightly to fasten it. Traditional instructions are printed sheets or a monitor next to the station. The worker has to read, remember, look back at the product and match what was read with what is seen. This takes time and leads to mistakes, especially when product variants change often.

1.2 Problem statement

Design an Adaptive AR Work Instruction system for machine assembly lines that overlays step by step guidance on the real machine part and automatically adjusts the level of guidance to the skill of the operator.

1.3 Augmented Reality in manufacturing

Augmented Reality adds computer generated information to the real scene in real time and registers it with real objects. In assembly, AR can show the exact bolt to tighten, the position of a part that is not yet fitted, and the next tool to use. Compared with Virtual Reality, the worker still sees the real station, which is essential for safety and quality.

Feature	Paper / screen manual	AR work instruction
Where information is shown	Separate sheet or monitor	On the real part, in the field of view
Finding the right step	Manual page turning	Automatic, step follows the work
Part and bolt identification	Picture and part number	Highlighted part, arrow and ghost image
Error detection	Final inspection	Immediately at the step (vision + sensors)
New product variant	Reprint and retrain	Publish new instructions to devices
Matching worker skill	Same for everyone	Adaptive detail level

1.4 Need for adaptive instructions

A new operator needs animations, text and warnings for every step. An expert finds the same content slow and distracting, and starts to ignore the screen, which also hides important safety messages. An adaptive system gives each operator the right amount of help and brings the detail back when the operator makes errors.

1.5 Scope of the project

- One gearbox assembly station of nine steps with four fastened bolts on the bearing cover.
- Three skill levels (novice, skilled, expert) and automatic switching between them.
- Authoring in a web studio, display on an AR tablet / headset, supervisor dashboard.
- The screenshots in this report are simulated screens built to show the design; no factory trial was run.

1.6 Benefits expected

Benefit	How it is achieved
Shorter training	Full guidance for novices, no classroom for each variant
Fewer errors	Vision and torque checks at every step with tool lock
Faster cycle	Hands-free voice control and automatic step advance
Safety	Warnings shown at the point of risk, in the line of sight
Data	Step times and errors stored for continuous improvement

1.7 VR, AR and MR compared

Technology	Real world visible?	Typical device	Use in assembly
Virtual Reality (VR)	No, fully virtual	Closed headset	Training before going to the line, planning
Augmented Reality (AR)	Yes, with overlays	Tablet, smart glasses	Live guidance on the real part (this project)
Mixed Reality (MR)	Yes, virtual objects interact	HoloLens 2	Guidance with spatial anchors and remote help

1.8 Challenges of today's assembly instructions

- **Information overload:** long manuals hide the few facts that matter for the current step.
- **Frequent variants:** each change of product needs new printed sheets and retraining.
- **Skill gap:** new operators and experts are given the same instruction.
- **Late detection:** a wrong bolt is usually found at the final inspection, after more work is added.

1.9 Applications

- Automobile and engine assembly lines.
- Electrical panel and wiring harness assembly.
- Aircraft and machinery maintenance and repair.
- Training of new workers and cross-training between stations.

1.10 Existing AR work instruction solutions

Solution	Main idea	Gap addressed here
Microsoft Dynamics 365 Guides	Step by step holographic guidance on HoloLens	Detail level is authored, not tuned to each worker
PTC Vuforia work instructions	AR authoring from CAD with tracking of the product	Limited use of live skill data
Connected worker platforms	Digital instructions with skills tracking and analytics	Guidance not linked to vision and tool checks per step

2. DESCRIPTION

2.1 System overview

The system has three parts: the shop-floor devices (AR device, torque tool, scanner, PLC), the adaptive work instruction server, and the web studio used by engineers and supervisors. The server holds the instructions, the operator profiles and the rules. The AR device only has to track the part, render the overlay and send camera frames and sensor events.

Module	Function
Tracking service	Recognises the gearbox from its CAD model (Model Target) and keeps the overlay fixed to it; QR marker as fallback
Instruction database	Stores steps, 3D models, animations, texts, audio and safety notes for each product variant
Step manager	State machine that moves from step to step only when the completion checks are passed
Vision checker	CNN that checks part type, bolt length, orientation and presence of grease or gasket
Torque tool bridge	Receives torque and angle values over MQTT and locks the tool on a wrong step
Adaptation engine	Calculates the skill score and selects the detail level for every step
Operator profile service	Keeps skill score, history, language and preferred interaction (voice / touch)
Analytics dashboard	Shows cycle time, errors by step, first pass yield and station status

2.2 Overlay elements used

Element	Meaning	Colour
Numbered ring	Fastener or location to work on, number gives the order	Green
Tick on grey ring	Fastener already completed and confirmed	Grey
Red ring and banner	Wrong part or failed check, work is held	Red
Ghost part	Transparent copy of the part that must be fitted	Blue
Arrow / line	Direction of movement or location of the correct bin	Blue
Instruction card	Title, tool, torque, text and buttons	Dark blue

2.3 Adaptation of the instruction

After every step the adaptation engine updates a skill score between 0 and 1 for the operator. The score rises when steps are finished within the target time without errors and falls when time is exceeded by more than 20% or errors occur. The score selects the detail level of the next steps.

Level	Skill score	What the operator sees
Novice	below 0.40	3D animation, three text lines, safety warning, voice help, confirmation of each step
Skilled	0.40 to 0.75	Highlights, torque value, short text; warnings only for risky steps
Expert	0.75 and above	Bolt highlight and torque value only; step advances on the sensor signal

2.4 Rules of the adaptation engine

Condition	Action
Step time more than 20% above target	Reduce skill score; keep or increase detail
Two errors in the last five steps	Step down one detail level and notify the supervisor
Five consecutive steps on time and without error	Step up one detail level
Risky step (torque, hot part, moving part)	Always show the safety card, even for experts
New or changed product variant	Show one level more detail for the first three units
Operator asks for help by voice	Show full detail for this step only

2.5 Information exchanged during one step

From	To	Data
Step manager	AR device	Step id, detail level, 3D assets, torque target
AR device camera	Vision checker	Camera frame (10 per second)
Vision checker	Step manager	Part / bolt result with confidence value
Torque tool	Torque bridge (MQTT)	Torque value, angle, tool id
Step manager	Adaptation engine	Step time and error count
Adaptation engine	Profile service	Updated skill score of the operator
Server	Dashboard	Station, step, cycle time, errors

2.6 Safety and ergonomics

- **Safety first:** a safety card is always shown for torque, hot and moving-part steps, at every level.
- **Clear view:** overlays are semi-transparent and kept away from the area where the hands work.
- **Short sessions:** tablets on a stand or light glasses are used to avoid fatigue over a full shift.
- **Fail-safe:** if tracking is lost the overlay freezes and the torque tool is locked.

3. WORKING STEPS

3.1 Steps followed

- Step 1:** **Study the assembly:** list the nine steps of the gearbox GB-220 station, the parts, tools, torque values and risks.
- Step 2:** **Prepare the 3D data:** import the CAD model, reduce it for mobile use and create the Model Target used for tracking.
- Step 3:** **Author the instructions:** in the web studio write each step with title, tool, torque, completion check and three detail levels.
- Step 4:** **Define the adaptation rules:** set the skill thresholds (0.40 and 0.75), the 20% time tolerance and the error rules.
- Step 5:** **Build the AR application:** create the Unity scene with the Model Target, instruction cards, highlights, ghost part and arrows.
- Step 6:** **Connect the tools:** link the torque wrench and scanner through MQTT and the vision checker to the camera feed.
- Step 7:** **Start the work order:** the operator scans the order QR code; the profile service returns the skill score.
- Step 8:** **Run the steps:** the overlay shows each step at the selected detail level; the step ends when the sensor and vision checks pass.
- Step 9:** **Handle errors:** a wrong part turns the ring red, locks the tool and informs the supervisor until the check is passed.
- Step 10:** **Adapt and log:** after each step the skill score is updated; times and errors are sent to the dashboard.

Output 1: Instruction authoring in the web studio (browser)

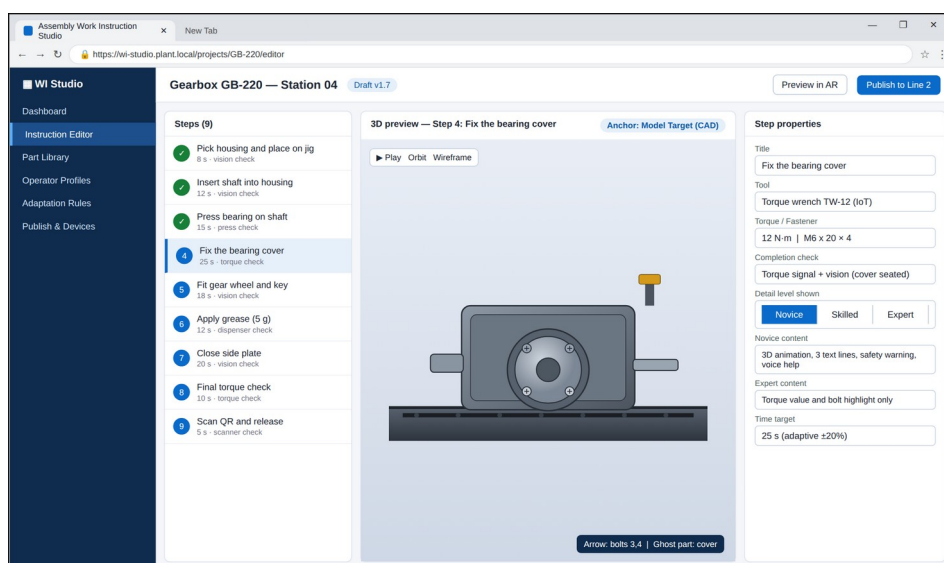


Fig. 1 - Step 4 "Fix the bearing cover" with 3D preview, tool, torque and the three detail levels.

Output 2: AR view for a novice operator

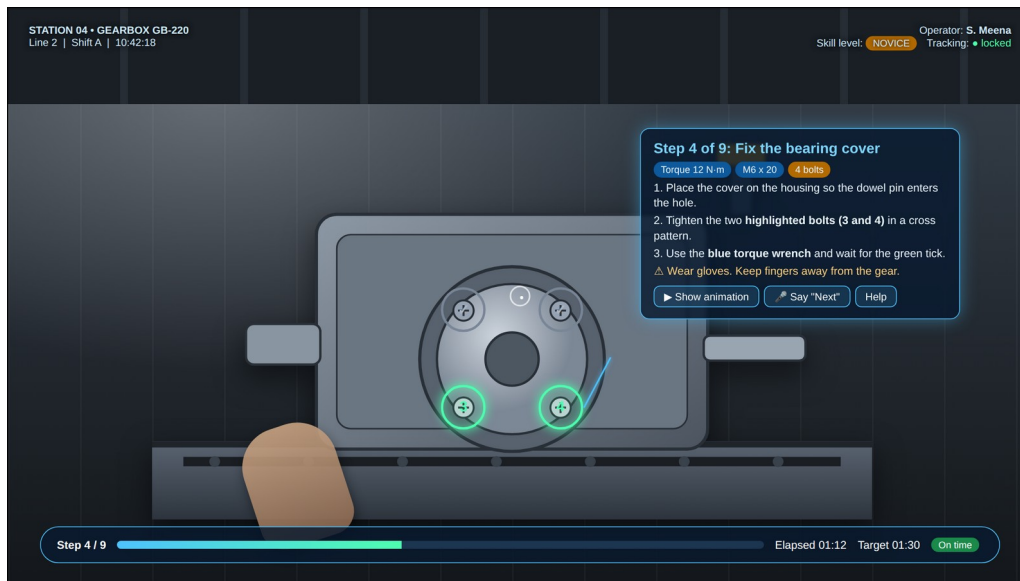


Fig. 2 - Highlighted bolts 3 and 4, animation button and full text for Step 4 of 9.

3.2 Operator view at different skill levels

Output 3: AR view for an expert operator

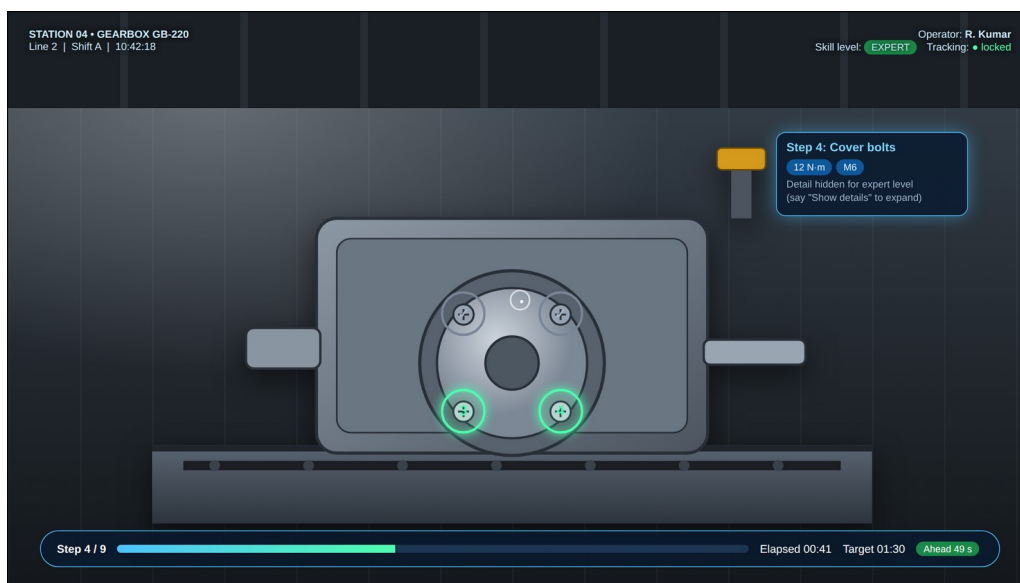


Fig. 3 - Only the bolt rings and the torque value are shown; the card is small and Step 4 is ahead of time.

Output 4: Error detected by the vision checker

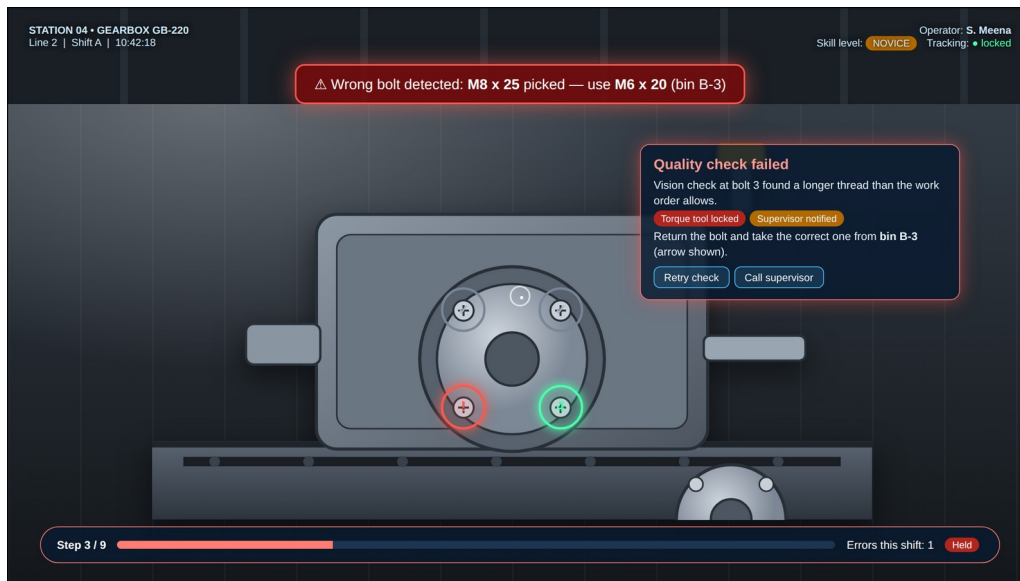


Fig. 4 - A wrong bolt (M8) is detected at bolt 3; the ring turns red, the tool is locked and the supervisor is notified.

3.3 Handheld (mobile) view

Output 5: Instruction on a mobile phone / tablet

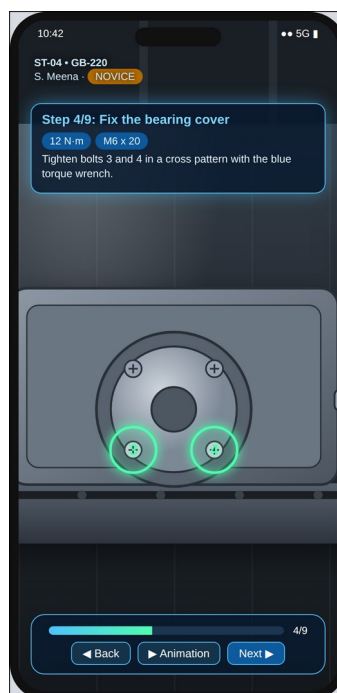


Fig. 5 - Compact card, step progress and Back / Animation / Next buttons.

3.4 Gearbox GB-220 assembly steps and checks

Step	Operation	Tool / part	Check	Target (s)
1	Pick housing and place on jig	Housing H-220	Vision: part and position	8
2	Insert shaft into housing	Shaft S-14	Vision: seated	12
3	Press bearing on shaft	Press, bearing B-6204	Press force signal	15
4	Fix the bearing cover	Torque wrench 12 N·m, M6 x 20 x 4	Torque + vision	25
5	Fit gear wheel and key	Gear G-40, key	Vision: key in slot	18
6	Apply grease (5 g)	Dispenser	Dispenser volume	12
7	Close side plate	Plate P-3, 6 bolts	Vision + torque	20
8	Final torque check	Torque wrench	Torque sweep	10
9	Scan QR and release	Scanner	Order match	5

3.5 Voice commands used

Command	Action
"Next"	Go to the next step (when no sensor signal is available)
"Back"	Return to the previous step
"Show animation"	Play the 3D animation of the current step
"Show details" / "Hide details"	Increase or reduce the detail level for this step
"Call supervisor"	Send an alert to the supervisor dashboard

3.6 Sample session of one operator (simulated)

Time	Event	Detail level shown
10:40:55	Operator S. Meena scans work order WO-5521	Novice (score 0.31)
10:41:20	Steps 1 and 2 finished within target time	Novice
10:41:49	Step 3 took 21 s against 15 s target	Novice (score 0.29)
10:42:11	Wrong bolt M8 detected at step 4, tool locked	Novice, error banner
10:42:30	Correct bolt fitted, torque 12.1 N·m accepted	Novice
11:30:00	After 5 correct steps in a row score rises to 0.44	Skilled

4. DIAGRAMMATIC REPRESENTATION

4.1 System architecture

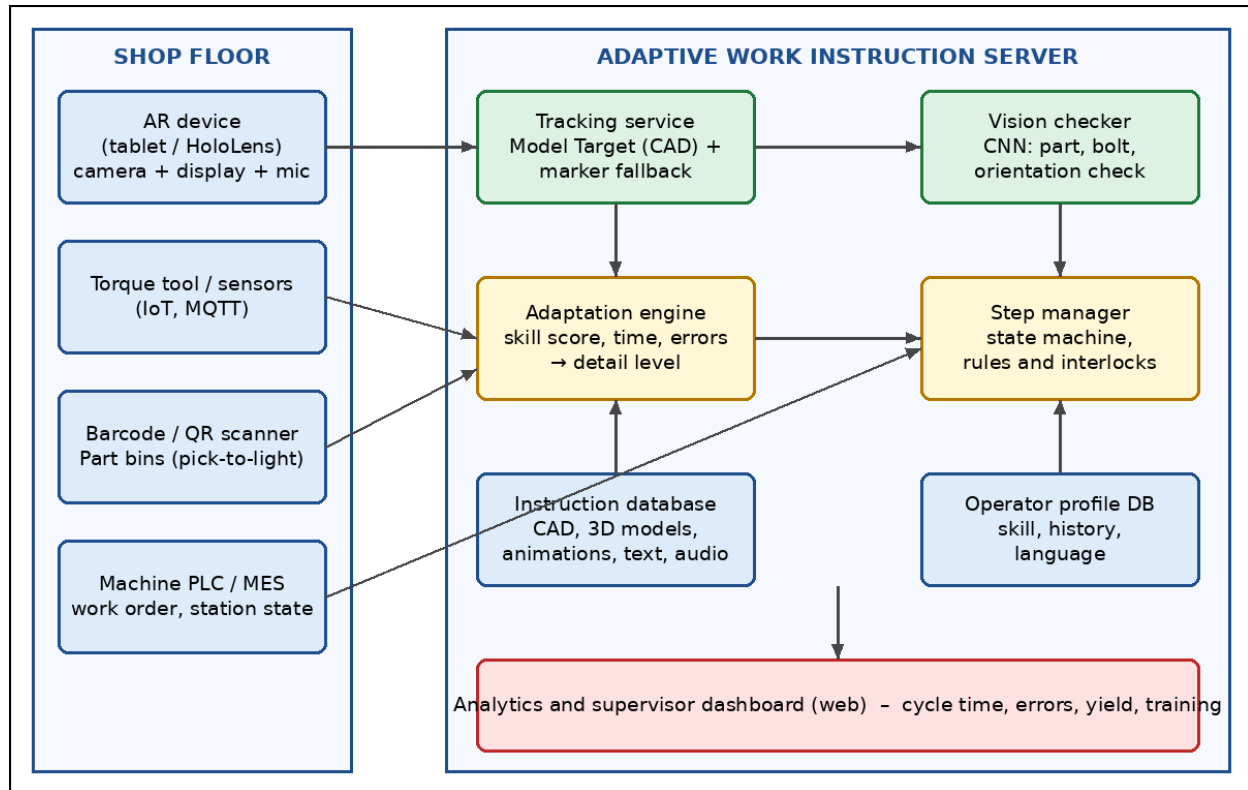


Fig. 6 - Shop-floor devices, the adaptive instruction server and the analytics dashboard.

4.2 Detail levels of the instruction

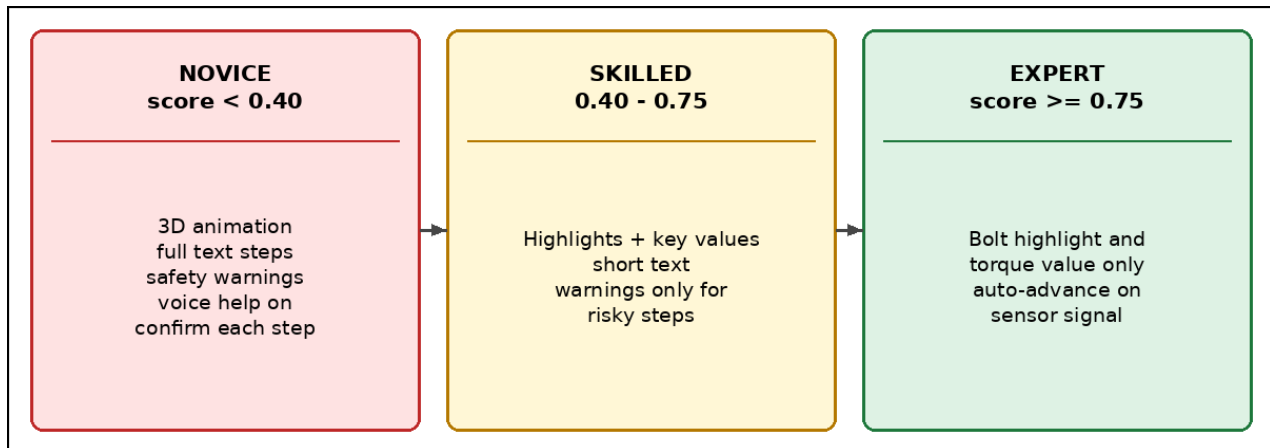


Fig. 7 - Novice, skilled and expert levels selected from the skill score.

4.3 Flowchart of the adaptive instruction process

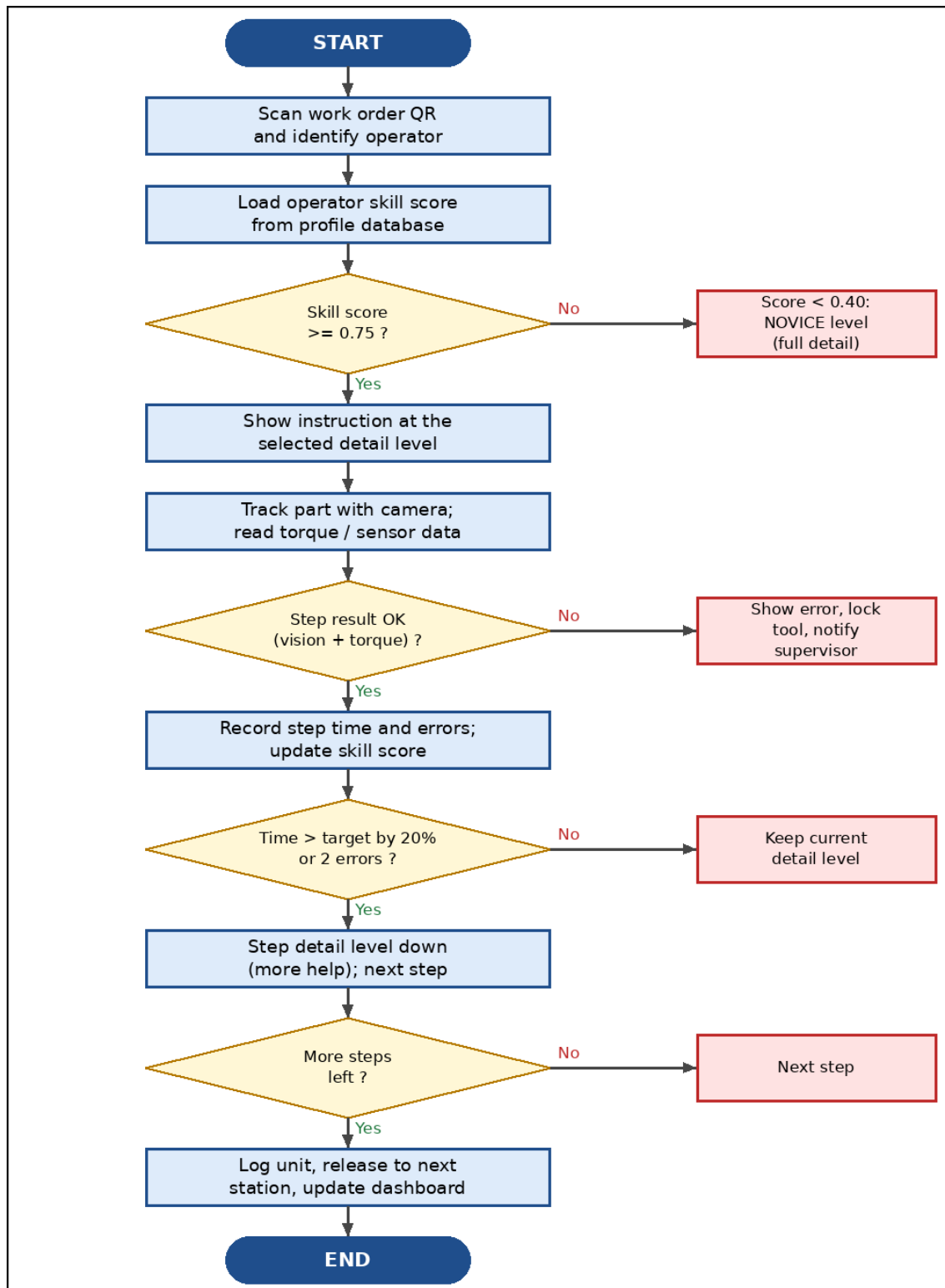


Fig. 8 - From the work order scan to the release of the unit with adaptation after every step.

4.4 Supervisor dashboard (sample data)

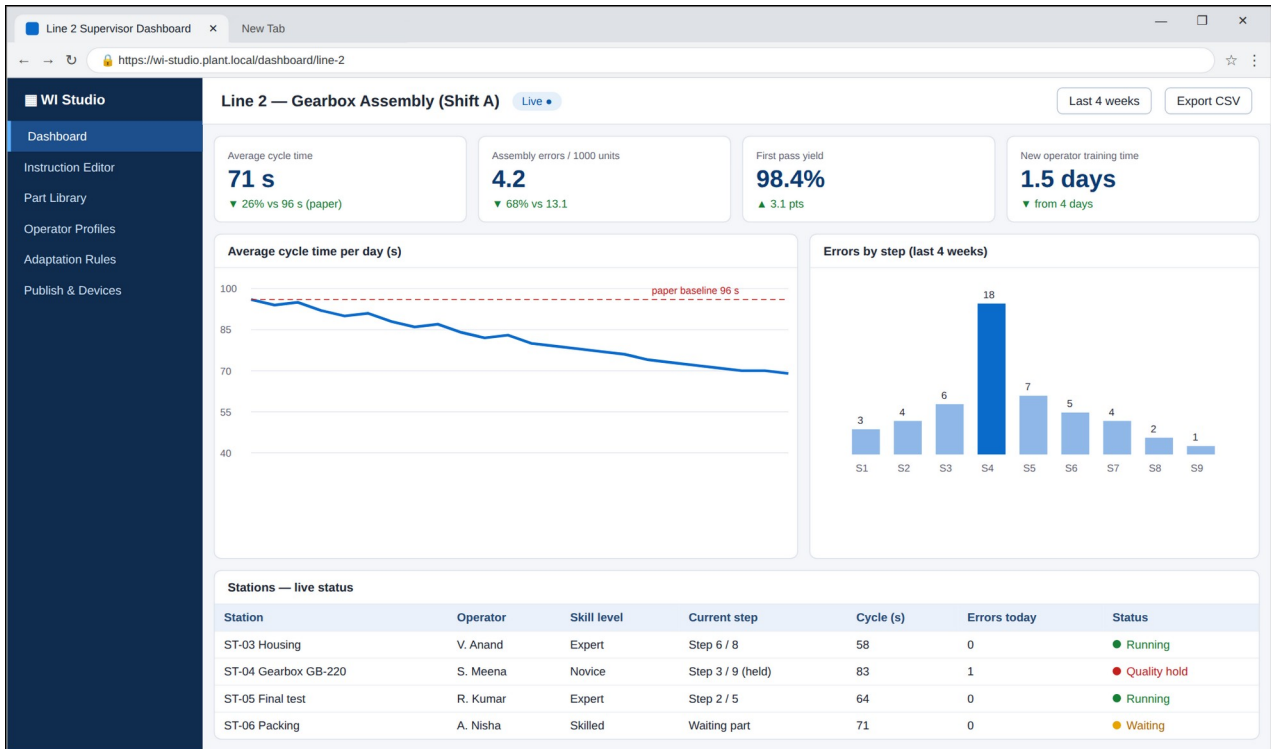


Fig. 9 - Dashboard of Line 2 showing cycle time, errors by step and live station status. The values are sample data of the simulation, not measurements from a factory.

4.5 Development environment

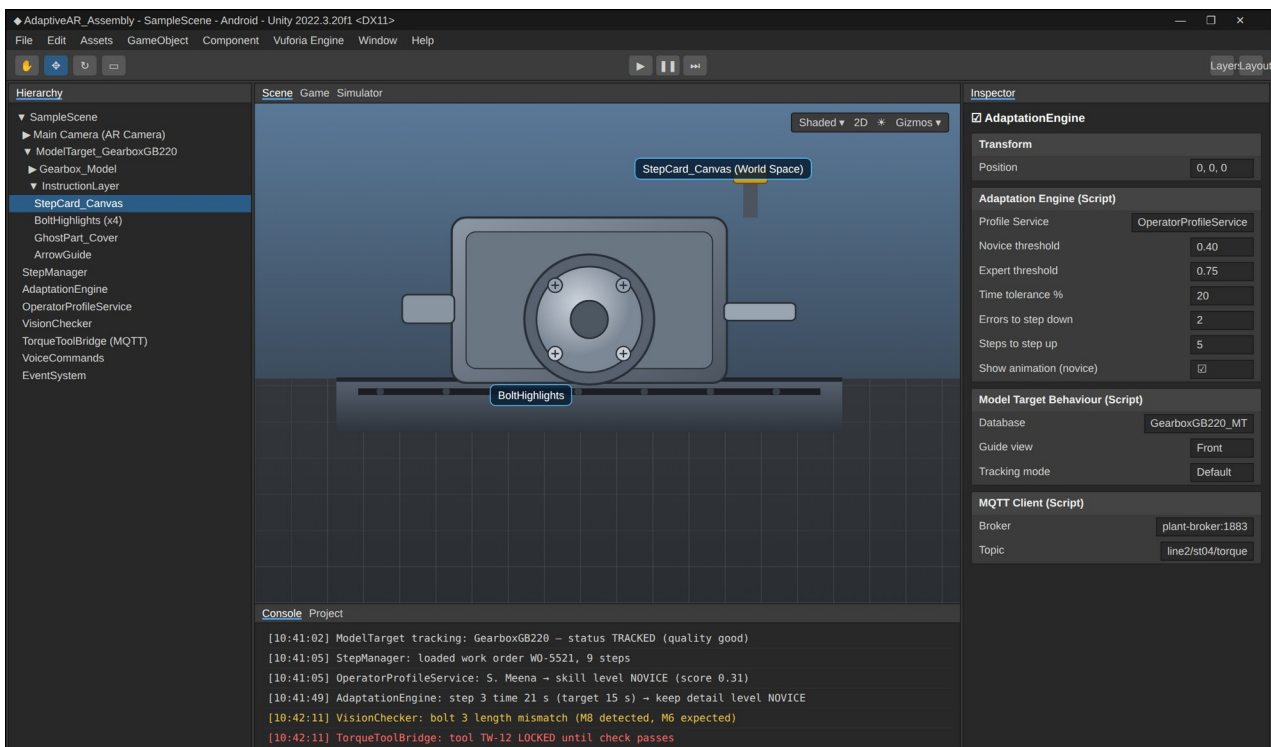


Fig. 10 - Unity scene with the Model Target, the instruction layer, the Adaptation Engine settings and the console log of one step.

5. CONCLUSION

An Adaptive AR work instruction system for a machine assembly line was designed. The instructions are overlaid on the real gearbox so the operator does not need to look away, and they are checked by sensors and a vision model at every step. The adaptation engine changes the detail level from the operator skill score, giving animations and warnings to new workers and a quiet, fast view to experts. The authoring studio, the AR operator views, the error handling, the dashboard and the Unity setup were presented as simulated screens. A real trial on a line is needed to measure the actual reduction in training time, cycle time and errors.

Advantages

- Guidance is in the line of sight and fixed to the part.
- Errors are stopped at the moment they happen.
- Each operator gets the amount of help that is needed.
- Instructions for new variants are published digitally in minutes.

Limitations

- Tracking can fail with reflective parts, poor light or when the part is covered by the hands.
- A head-mounted display can be tiring for a full shift; tablets or lighter glasses may be preferred.
- The skill score depends on enough data; the first units of a new worker use the novice level.

Future enhancements

- Hand tracking to detect which part the worker picks, with pick-to-light bins.
- Learning the best detail level for each step with machine learning.
- Remote expert support with a shared AR view for rare faults.
- Digital twin of the line to test new variants before release.
- Multi-language voice and text for a mixed workforce.

References

S. No.	Reference
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2	PTC Vuforia Engine, Model Targets developer guide
3	Microsoft, HoloLens 2 and Dynamics 365 Guides documentation
4	ISO 9241-210, Human-centred design for interactive systems
5	Azuma R., A Survey of Augmented Reality, Presence: Teleoperators and Virtual Environments, 1997